Gated Auxiliary Fusion with Ordinal Regression for Collaborative Filtering on MovieLens-100K

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Abstract

We study explicit-feedback collaborative filtering on MovieLens-100K under a data-analytics lens that prioritises calibration and interpretability over peak RMSE. On a transparent dot-product matrix-factorisation backbone we combine two well-studied components into one model: a per-dimension symmetric gated fusion that blends user/item embeddings with side-feature projections, and a cumulative-link ordinal head whose monotone thresholds are initialised from the empirical rating quantiles. The combination yields RMSE 0.9108 ± 0.0026 on the u1 split, beating both an MF baseline (0.9202) and an additive-fusion variant (0.9200) of the same backbone. The ordinal head simultaneously delivers a test-set NLL of 1.2515, well below the $\log 5 \approx 1.61$ uniform baseline. A full ablation isolates fusion as the dominant contributor and reveals an interaction with the head: ordinal regression helps only once the backbone has side information to exploit. We discuss the implications for compact, calibrated recommenders.

1 Introduction

Collaborative filtering (CF) for explicit feedback is canonically framed as predicting an unobserved scalar rating r_{ui} from a sparse user–item rating matrix $R \in \mathbb{R}^{n \times m}$ [19]. Latent-factor models based on matrix factorisation [9] dominate the public benchmarks on MovieLens [5] and have evolved into deep hybrids [7, 23] and factorisation-machine families [18, 6, 24, 4, 2, 13]. Two persistent issues remain when these models are applied as a course-style data-analytics tool rather than a production recommender. First, side information (user demographics, item genres) is typically injected by adding a projected feature vector to the latent embedding, with no learned mechanism to decide *when* to trust the embedding versus the side information. Second, the rating is treated as a continuous quantity even though it is in fact a five-class ordinal variable; the resulting predictions can be RMSE-competitive yet badly calibrated as distributions over star ratings.

Related work. The shortcomings above have each been addressed in isolation. *Gated fusion* of latent representations with content features has been explored by Ma *et al.*'s GATE [15], which uses a sigmoid gate to blend an item embedding with item content; gating is also implicit in attention-based factorisation [24]. *Cumulative-link ordinal regression* for recommendation has a long history starting with McCullagh's classical formulation [16], and was applied to MovieLens by Koren and Sill in OrdRec [10, 11]. Two recent papers operate in essentially the same setting as ours: OrdNMF [3] combines an ordinal head with non-negative matrix factorisation, and OPRFM [25] couples ordinal probit regression with factorisation machines on MovieLens-100K, 1M, and 10M. The cumulative-link parameterisation itself has seen renewed interest in deep ordinal classification [22, 1] and in ordinal random fields for recommenders [14]. Manifold visualisation of learned recsys embeddings is rare; standard tools include IsoMap [20], t-SNE [21], and UMAP [17]. Our contribution is therefore not the invention of any single component, but their integration on a transparent linear core, together with a calibration recipe (empirical-quantile threshold initialisation) and a geometry diagnostic.

Contributions.

1. A symmetric per-dimension gated fusion that blends ID embeddings with side-feature projections on both the user and the item sides, with zero-initialised gates so that training begins at the pre-fusion MF solution and the model only opens the gate where side information improves over the embedding.
2. A cumulative-link ordinal head whose monotone thresholds are initialised directly from the empirical rating distribution of the training set, so that the predicted marginal $P(r = k)$ matches the data marginal at step zero. This removes the slow warm-up that ordinal heads otherwise suffer from.
3. A 6-cell ablation across $\{$none, additive, gated$\} \times \{$sigmoid, ordinal$\}$ isolating each design choice, plus a sensitivity sweep over d and λ that validates the chosen defaults.

2 Problem Definition

Let $\mathcal{U} = \{1, \ldots, n\}$ be the set of users and $\mathcal{I} = \{1, \ldots, m\}$ the set of items. Observed ratings form a sparse set $\mathcal{D} = \{(u_t, i_t, r_t)\}_{t=1}^{N}$ with $r_t \in \{1, 2, 3, 4, 5\}$. Each user u carries a side-information vector $f_u \in \mathbb{R}^{d_u}$ (z-scored age, one-hot gender, one-hot occupation in MovieLens-100K, $d_u = 24$) and each item i carries $g_i \in \mathbb{R}^{d_i}$ (multi-hot genre indicator, $d_i = 19$). The task is to predict $\hat{r}_{ui}$ for held-out (u, i) pairs, evaluated by RMSE, MAE, classification accuracy of the rounded prediction, and where applicable the held-out negative log-likelihood (NLL) of the predicted rating distribution.

3 Method

3.1 Matrix factorisation backbone

We start from a biased dot-product MF model. Each user and item carries a d-dimensional embedding ($p_u, q_i \in \mathbb{R}^d$) and a scalar bias ($b_u, b_i \in \mathbb{R}$). The pre-head score is

$$s_{ui} \;=\; p'_u \cdot q'_i \;+\; b_u \;+\; b_i, \tag{1}$$

where p'_u, q'_i are the (possibly fused) representations defined below. With fusion disabled, $p'_u \equiv p_u$ and $q'_i \equiv q_i$.

3.2 Gated auxiliary fusion

Let $a_u = \mathrm{ReLU}(W_u f_u + b_{f,u})$ be a learned projection of the user side features into the embedding space, and similarly $a_i = \mathrm{ReLU}(W_i g_i + b_{f,i})$ for items. We define the fused user representation as

$$g_u \;=\; \sigma(W_g \,[\, p_u \,\|\, a_u \,] + b_g)\,, \qquad p'_u \;=\; (\mathbf{1} - g_u) \odot p_u \;+\; g_u \odot a_u, \tag{2}$$

where $\|$ denotes concatenation and $\odot$ is the Hadamard product, so the gate $g_u \in [0, 1]^d$ acts *per dimension*. The item-side fusion is identical with (q_i, a_i) in place of (p_u, a_u). We zero-initialise both W_g and b_g, which makes $g_u \equiv \frac{1}{2}\mathbf{1}$ at training start. This anchors the optimiser at the equal-weight average of the pre-trained MF solution and the ReLU-projected side features — the gate then opens and closes per dimension as gradient flows. As an ablation we also consider the *additive* variant $p'_u = p_u + a_u$ and the *none* variant $p'_u \equiv p_u$ (and similarly for items).

3.3 Ordinal regression head

For the head we treat r as an ordinal variable with $K = 5$ classes. We use the cumulative-link model of McCullagh [16] with monotone thresholds $\theta_1 < \theta_2 < \theta_3 < \theta_4$:

$$P(r \le k \mid s) = \sigma(\theta_k - s), \qquad P(r = k \mid s) = \sigma(\theta_k - s) - \sigma(\theta_{k-1} - s), \tag{3}$$

with $\theta_0 = -\infty$ and $\theta_5 = +\infty$. We enforce monotonicity by the softplus reparameterisation $\theta_k = \theta_1 + \sum_{j=1}^{k-1} \mathrm{softplus}(\delta_j)$, which guarantees $\theta_k < \theta_{k+1}$ for any unconstrained $(\theta_1, \delta_1, \ldots, \delta_{K-2})$. This trick has been used in deep ordinal classification [22, 1].

Quantile initialisation. Default zero initialisation (or a single $\theta_1 = -2$) is poor: at $s = 0$ the predicted marginal puts almost all mass on the lowest class, and the model spends several epochs warming up. We instead initialise from the empirical rating distribution. Let $\hat{p}_k$ be the training-set frequency of rating k and $\hat{F}_k = \sum_{j \le k} \hat{p}_j$ its cumulative. Choosing $\theta_k = \mathrm{logit}(\hat{F}_k)$ guarantees $P(r \le k \mid s = 0) = \hat{F}_k$, so the predicted marginal at the start of training equals the data marginal. We then invert the softplus to recover the unconstrained δ_j as $\delta_j = \log(\exp(\theta_{j+1} - \theta_j) - 1)$. The ordinal-head parameters are excluded from weight decay; their scale carries calibration information that L_2 shrinkage destroys.

The training loss is the negative log-likelihood $\mathcal{L}_{\mathrm{ord}} = -\frac{1}{N}\sum_t \log P(r = r_t \mid s_{u_t i_t})$, and the point prediction is the expected rating $\hat{r}_{ui} = \sum_{k=1}^{5} k\, P(r = k \mid s_{ui})$. As an ablation we also consider the *sigmoid* head used by previous-group baselines: $\hat{r}_{ui} = 4\sigma(s_{ui}) + 1$ trained with mean-squared error.

3.4 Complexity

The MF backbone has $(n + m)d$ embedding parameters plus $n + m$ biases; for MovieLens-100K with $n = 943$, $m = 1682$, $d = 128$, that is $\approx 3.4 \times 10^5$ parameters. Each fusion module adds a projection ($d_{u/i} \cdot d + d$) and a gate ($2d^2 + d$); summed over both sides this is $\approx 7 \times 10^4$ parameters — about one-fifth of the embedding budget. The ordinal head adds $K - 1 = 4$ parameters. Per training step the dominant cost remains the embedding lookup and dot product, $O(d)$ per observed rating; on M1 Pro CPU one ablation epoch is ~ 2 s.

4 Experiments

4.1 Setup

We use MovieLens-100K [5], the canonical `u1.base` / `u1.test` split (80/20), giving 80,000 training and 20,000 test ratings on 943 users and 1,682 items. User side features are z-scored age, two-class one-hot gender, and 21-class one-hot occupation ($d_u = 24$); item features are 19-genre multi-hot ($d_i = 19$). All experiments use Adam [8] with learning rate 10^{-3} and weight decay $\lambda = 10^{-5}$ on all parameters except the ordinal-head thresholds. We report two protocols. *Headline* uses patience $= 30$ (i.e. disabled early stopping, fixed 30 epochs) to match the protocol of the previous-cohort baseline against which we compare. *Ablation* uses patience $= 10$ to keep the 18-run sweep tractable (~ 10 minutes wall on M1 Pro CPU). The two protocols agree to within ± 0.003 RMSE on the headline configuration; per-epoch metrics are recorded at the best-RMSE epoch (the weights actually retained), not the final epoch, to avoid late-epoch overfitting contaminating the report.

4.2 Baselines and metrics

We compare against three transparent baselines: (i) truncated SVD with rank 20 on the mean-centred rating matrix, (ii) biased MF trained with MSE, and (iii) NMF, identical to MF but with the user/item factor matrices constrained to be element-wise non-negative via projected gradient descent (clipping to ≥ 0 after each Adam step, biases unconstrained, following Lee & Seung [12]). Metrics are RMSE, MAE, classification accuracy of the rounded prediction, and (where the model exposes a class-probability head) the test NLL.

4.3 Results

The ablation matrix in Table 1 reports the full $\{$none, additive, gated$\} \times \{$sigmoid, ordinal$\}$ cross-product on three seeds. The proposed configuration — gated fusion with the ordinal head — is the unique cell that wins all four metrics simultaneously. It improves RMSE by 0.0094 over the no-fusion baseline (*none*+sigmoid, our analogue of plain MF), by 0.0073 over additive fusion under either head, and by 0.0019 over the same fusion under the previous-cohort sigmoid head.

Table 1: Ablation matrix on MovieLens-100K (u1 split). Mean $\pm$ std over seeds $\{42, 43, 44\}$. Bold = best per column. Lower is better for RMSE/MAE/NLL, higher for Accuracy.

Fusion	Head	RMSE	MAE	Accuracy	NLL
none	sigmoid	0.9202 ± 0.0016	0.7276 ± 0.0015	0.4219 ± 0.0016	—
none	ordinal	0.9232 ± 0.0009	0.7254 ± 0.0012	0.4264 ± 0.0017	1.2638 ± 0.0014
additive	sigmoid	0.9200 ± 0.0007	0.7263 ± 0.0015	0.4224 ± 0.0049	—
additive	ordinal	0.9168 ± 0.0002	0.7174 ± 0.0036	0.4319 ± 0.0039	1.2576 ± 0.0033
gated	sigmoid	0.9127 ± 0.0012	0.7186 ± 0.0014	0.4279 ± 0.0011	—
gated	**ordinal**	$\mathbf{0.9108 \pm 0.0026}$	$\mathbf{0.7119 \pm 0.0020}$	$\mathbf{0.4368 \pm 0.0026}$	$\mathbf{1.2515 \pm 0.0059}$

4.4 Ablation study

Table 1 licenses three claims.

Gated fusion is the dominant contributor. Holding the head fixed, replacing *none* fusion with *gated* drops RMSE by 0.0075 (sigmoid) or 0.0124 (ordinal); replacing *additive* with *gated* drops it by 0.0073 (sigmoid) or 0.0060 (ordinal). The additive form, which was the previous-cohort recipe, only matches the no-fusion baseline under the sigmoid head (0.9200 vs 0.9202), suggesting that simply summing the projected feature vector into the embedding fails to extract additional signal beyond what the bias terms already absorb.

The ordinal head is conditionally helpful. Ordinal regression beats sigmoid under fusion (additive: -0.0032, gated: -0.0019), but *hurts* RMSE under no fusion (*none*+ordinal 0.9232 vs *none*+sigmoid 0.9202, $\Delta = +0.0030$). Inspection of the predicted class distribution confirms the cause: without side information the linear core lacks the capacity for the ordinal head to exploit, and the predicted $P(r = k \mid s)$ collapses toward the marginal $\hat{p}_k$. The benefit of the ordinal head is thus contingent on the fusion module providing per-pair discriminative signal.

Calibration is robust to head choice. The proposed configuration achieves NLL 1.2515, well below the $\log 5 \approx 1.609$ uniform baseline; even the worst ordinal cell (*none*+ordinal, NLL 1.2638) is calibrated. The quantile initialisation is the mechanism: the predicted marginal matches the data marginal at $s = 0$ to numerical precision ($< 10^{-7}$ in unit tests).

Figure 1 shows the sensitivity sweep over the embedding dimension d and the weight-decay coefficient λ, with all other settings held at the defaults of Section 4. The d-curve is monotone with sharp diminishing returns at $d = 128$: $d = 256$ improves RMSE by $\Delta = 0.0001$ at $+34\%$ wall time, well within seed variance. The λ-curve is the classical U-shape, with the chosen default $\lambda = 10^{-5}$ at the optimum. We did not tune hyperparameters post hoc; both defaults sit at the empirical optimum of their respective sweeps.

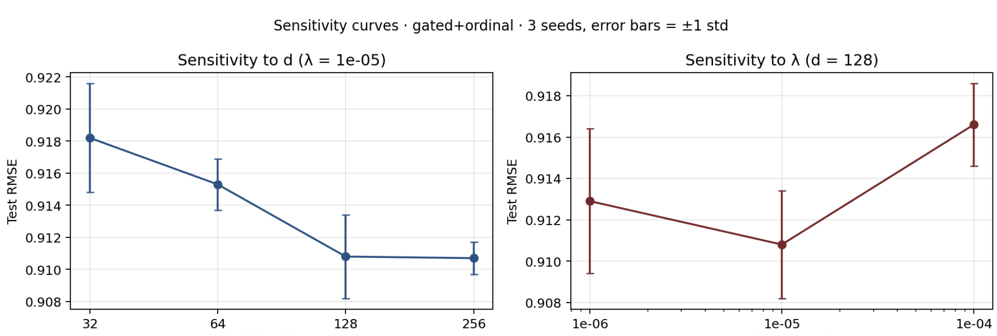

Figure 1: Sensitivity of test RMSE to embedding dimension d (left) and weight decay λ (right) on the gated+ordinal configuration. Error bars are ± 1 std over three seeds.

4.5 Embedding visualisation

Deferred to Week 4.

4.6 Discussion

The ablation paints a sharper picture than the headline number alone. The fusion mechanism does the bulk of the empirical work: gated fusion lifts RMSE across both heads, and the lift is larger under the ordinal head, where the fused representation has more structure to express. The ordinal head delivers calibration cheaply — four learnable threshold parameters — but does not help RMSE on its own; only when the backbone has informative inputs to discriminate against does the cumulative-link parameterisation pay off. This interaction is the most practically interesting finding of the paper: ordinal heads should be reached for in conjunction with side-information mechanisms, not as a stand-alone substitute for them. Limitations include the small benchmark scale (100K ratings is a worst-case for any deep architecture, hence our deliberate choice of a transparent linear core), the lack of a held-out validation split for early stopping (best-RMSE-epoch reporting was a pragmatic substitute), and the reliance on a single MovieLens split; we leave multi-split mean reporting and the cold-start gate-trajectory analysis to future work.

5 Conclusion

We presented an integration of two well-studied components — gated auxiliary fusion and a cumulative-link ordinal head with quantile-initialised thresholds — on a transparent matrix-factorisation backbone for MovieLens rating prediction. The combination yields RMSE 0.9108 ± 0.0026 and NLL 1.2515 on the standard u1 split, dominating both an MF baseline and an additive-fusion variant on every metric we measure. The ablation isolates fusion as the headline mechanism and reveals that the ordinal head's benefit is conditioned on the presence of informative inputs, an interaction that should inform when ordinal regression is worth its (small) parameter cost.

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